

PRATYUSH MAJUMDAR

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EDUCATION

MES Institute of Management and Career Courses
MCA (Pursuing) – September 2024 - April 2026

Mayur Colony, Pune
CGPA: 9.62

MES Garware College of Commerce
BCA – September 2021 - April 2024

Karve Road, Pune
CGPA: 9.45

EXPERIENCE

MES Institute of Management and Career Courses

Mayur Colony, Pune

Signature Forgery Detection - Machine Learning Researcher

June 28, 2025 - Present

- Built machine learning models such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), achieving classification accuracies of 66.4% (SVM) and 62.2% (KNN) for detecting forged signatures.
- Applied advanced image processing techniques (e.g., Gaussian blur, edge detection, thresholding) and extracted features using Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP), improving model input quality and boosting classification performance by 15%.

PROJECTS

AI Assistant - AI Developer

- Developed an AI Assistant using Python, Streamlit, and LangChain, enabling users to interact with an intelligent conversational agent for various tasks, improving task efficiency by 25%.
- Implemented vector embeddings using Hugging Face Embeddings and LangGraph along with custom tools to enhance the chatbot's ability to manage complex interactions, boosting conversation handling capabilities by 30%, while resolving the user queries and streamline the process.
- Designed and implemented a CI/CD pipeline using Jenkins, Docker, SonarQube and Nexus and deployed the application on Azure Kubernetes Service (AKS) to ensure automated build, testing, and deployment.
- [Live Demo](#) | [GitHub Repository](#)

TaxQueryAI - Generative AI Engineer

- Engineered TaxQueryAI, a MySQL database Q&A tool using Python, Next.js, and LangChain framework, enabling users to query tax-related data effortlessly through natural language, boosting query efficiency by 25%.
- Leveraged MySQL to dynamically generate queries from the database and applied NLP techniques to transform outputs into humanized responses, enhancing user comprehension and satisfaction by 30%.
- Integrated LangChain framework for advanced language understanding, ensuring accurate and contextually relevant SQL query generation, improving efficiency by 25%.
- [Live Site](#) | [GitHub Repository](#)

TECHNICAL SKILLS

Programming: Python, MySQL, Linux Shell Scripting, Java

AI/ML: Generative AI, NLP, Machine Learning, LangChain, LangGraph, Vector Embedding, Semantic Search

Tools: Power BI, Excel, Git, Jupyter Notebook, Docker, Jenkins, SonarQube, Sonatype Nexus

Cloud/Data Engineering: Databricks, Azure Data Factory, Azure Data Lake, Kubernetes

CERTIFICATIONS

Lets Talk Academy – [N3, JLPT](#) – [N4, JLPT](#) – [N5](#), Coursera – [Python for Data Science, AI and Development](#),

Coursera – [Data Analysis with Python](#), Udemy – [Azure Data Engineer Associate](#), Udemy – [Linux Shell Scripting](#)

ACHIEVEMENTS

- Second Prize in a Startup Idea Competition organized by SSMSIMR College.
- Assisted in a PhD project titled “English to Sanskrit Translator”.